Title

PAYMENT GATEWAY

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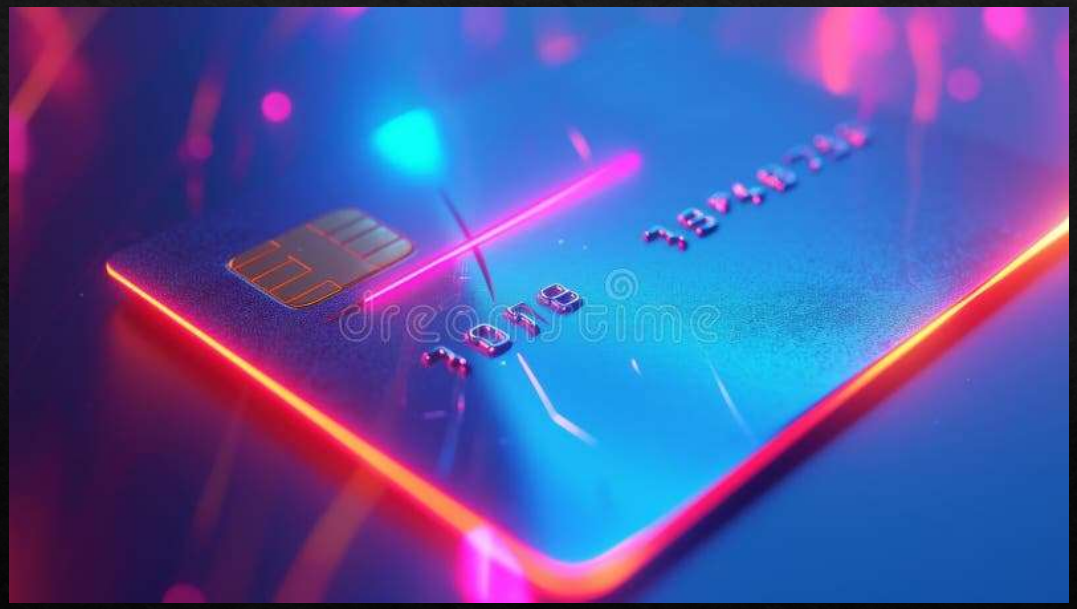
PAYMENT GATEWAY



INTRODUCTION

•A Payment Gateway is an e-commerce application service provider that authorizes payments for e-business, online retailers etc.

•The gateway itself is simply a secure online link between the merchant and the customer's bank. Sensitive information, such as credit card numbers, needs to be protected from fraudulent parties.

Steps Involved

- Customer initiates the purchase.
- Merchant collects the buyer information.
- Order is sent to the Payment Gateway.
- Payment Gateway collects the credit card information.
- Payment Gateway handles the transaction.
- Payment Gateway verifies the purchase.

PRELIMINARIES

•**Online Customer**: Entity who browses the e-commerce website and who will buy the products by making timely payments.

•**Merchant**: A merchant is the seller who will receive the payments made by the customer.

•**Banks**: There are two types of banks:
1.Client Bank: It holds the client's account and validates the customer.
2.Merchant Bank: It holds the merchant's account and is responsible for management and fraud control, etc.

•**Payment Gateway**: A payment gateway is connected to all the customers, merchants, and banks through the internet and is responsible for the speed, reliability, and security of all transactions that take place.

Framework Overview

Categories of Payment Gateways

- Two types:
  - Hosted Payment Gateways
  - Shared Payment Gateways
- **Hosted Payment Gateways:**
- Redirect users to external payment page
- Customer leaves merchant site temporarily
- No need to store sensitive data
- Examples: PayPal, WorldPay

Shared Payment Gateway

- Customer stays on merchant website
- Payment page opens within site
- Secure handling by gateway
- User returns after payment
- Fast and easy to use

Example: eWay

Security

- Is your money safe?
- Online payments are easy but involve risks
- Card details are shared with third parties
- Risk of online fraud exists
- Banks & platforms ensure secure transactions

Security (cntd...) – Data Encryption

- Protects payment data using encryption
- Card details are encrypted with public key
- Only gateway can decrypt using private key
-
- Prevents unauthorized access during transfer
- Uses symmetric or asymmetric cryptography

Security (cntd…) – SSL

- SSL creates a secure encrypted connection
- Protects data between browser and server
- Verifies website identity
- Secures card and personal details
- Look for **HTTPS & padlock** for safety

Security (cntd...) - SET

- Developed by Visa & MasterCard
- Secures online card transactions
-
- Hides card details from unauthorized users
- Uses dual signature for secure communication
- Ensures confidentiality and authentication

Types of Payment Gateways

- PayPal
- 2Checkout
- Amazon Payment
- Google Checkout
- SecurePay
- Authorize.Net
- PayUmoney

Google Checkout
SecurePay
Authorize.Net
PayUmoney
- PaySimple

Features of Payment Gateways

- Transaction & Network Security
- Secure with latest security technologies like SSL
- Fraud screening features
- Reduces risk of theft and loss
- User-friendly and speedy in processing
- Support for multiple forms of payment
- Time saving
- Ease of use

Disadvantages of Payment Gateways

- Payment gateways highly depend on internet connectivity
- Lack of anonymity
- Risk of being hacked
- Fees

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C: > Users > sanga > course project > payment gateway1.c > cardPayment(float)

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>

void processPayment(float amount) {
    printf("\nProcessing payment...\n");

    // simulate delay
    for(int i = 0; i < 1000000000; i++);

    srand(time(0));
    int success = rand() % 2;

    if(success) {
        printf("Payment Successful!\n");
        printf("Amount Paid: Rs. %.2f\n", amount);
    } else {
        printf("Payment Failed! Try again.\n");
    }
}

void upiPayment(float amount) {
    char upi[50];

    printf("Enter UPI ID: ");
    scanf("%s", upi);

    if(strchr(upi, '@') != NULL) {
        printf("Valid UPI ID\n");
        processPayment(amount);
    } else {
```

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```c
void upiPayment(float amount) {
    } else {
        printf("Invalid UPI ID\n");
    }
}

void cardPayment(float amount) {
    char card[20];
    char cvv[5];

    printf("Enter Card Number: ");
    scanf("%s", card);

    printf("Enter CVV: ");
    scanf("%s", cvv);

    if(strlen(card) == 16 && strlen(cvv) == 3) {
        printf("Card details valid\n");
        processPayment(amount);
    } else {
        printf("Invalid Card Details\n");
    }
}

int main() {
    int choice;
    float amount;

    printf("===== PAYMENT GATEWAY =====\n");
    printf("Enter Amount: Rs. ");
    scanf("%f", &amount);

```

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C: > Users > sanga > course project > payment gateway1.c > cardPayment(float)

```c

    printf("\nSelect Payment Method:\n");
    printf("1. UPI\n");
    printf("2. Card\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            upiPayment(amount);
            break;
        case 2:
            cardPayment(amount);
            break;
        default:
            printf("Invalid Choice\n");
    }

    return 0;
}
```

PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\sanga> cd 'c:\Users\sanga\course project\output'
PS C:\Users\sanga\course project\output> & .\'payment gateway1.exe'
===== PAYMENT GATEWAY =====
Enter Amount: Rs. 999

Select Payment Method:
1. UPI
2. Card
Enter choice: 1
Enter UPI ID: chandu@ybl
Valid UPI ID

Processing payment...
Payment Successful!
Amount Paid: Rs. 999.00
PS C:\Users\sanga\course project\output>
```

```
PS C:\Users\sanga\course project\output> & .\'payment gateway1.exe'
===== PAYMENT GATEWAY =====
Enter Amount: Rs. 666

Select Payment Method:
1. UPI
2. Card
Enter choice: 1
Enter UPI ID: gajajaj@eam
Valid UPI ID

Processing payment...
Payment Failed! Try again.
PS C:\Users\sanga\course project\output>
```

```
PS C:\Users\sanga\course project\output> & .\'payment gateway1.exe'
===== PAYMENT GATEWAY =====
Enter Amount: Rs.
8888

Select Payment Method:
1. UPI
2. Card
Enter choice: 2
Enter Card Number: 5898635614147895
Enter CVV: 323
Card details valid

Processing payment...
Payment Failed! Try again.
```

```
PS C:\Users\sanga\course project\output> & .\'payment gateway1.exe'
===== PAYMENT GATEWAY =====
Enter Amount: Rs. 3699

Select Payment Method:
1. UPI
2. Card
Enter choice: 2
Enter Card Number: 4047459764306358
Enter CVV: 072
Card details valid

Processing payment...
Payment Successful!
Amount Paid: Rs. 3699.00
```

Gracias